Heaven's Light is Our Guide

Rajshahi University of Engineering & Technology

Department of

Electrical & Computer Engineering

Project Report

SMS & Email Spam Detection System Using Naive Bayes · Logistic Regression · SVM with TF-IDF

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Abstract

Spam messages in SMS and email communications represent a persistent and growing problem imposing significant burdens on users and network infrastructure. This report presents the design, implementation, and evaluation of an automated spam detection system built using supervised machine learning techniques. The system uses the SMS Spam Collection dataset comprising 5,574 labelled messages (747 spam, 4,827 ham) from the UCI Machine Learning Repository. Text data is preprocessed through a pipeline involving lowercasing, URL removal, numeric removal, stopword filtering (NLTK), and Porter Stemming, after which Term Frequency-Inverse Document Frequency (TF-IDF) vectorisation produces a 5,000-dimensional feature space. Three classification models were trained and comparatively evaluated: Multinomial Naive Bayes, Logistic Regression, and a Linear Support Vector Machine (SVM). The best-performing model achieves accuracy exceeding 97% and F1-score above 0.96, with a ROC-AUC of approximately 0.99. The system is deployed as an interactive web application using Streamlit. Complexity analysis demonstrates Naive Bayes offers O(N·d) training time while SVM delivers higher accuracy at greater computational overhead. The project aligns with CO3 and addresses the accuracy-interpretability and speed-performance trade-offs (P2)

Keywords: Spam Detection, Naive Bayes, TF-IDF, SVM, Logistic Regression, NLP, Text Classification

1 Introduction

1.1 Background and Motivation

Electronic messaging has become one of the most pervasive forms of human communication. SMS and email together handle billions of messages daily. However, this accessibility has been exploited to deliver unsolicited commercial messages, phishing attempts, and fraudulent content — collectively known as spam. Approximately 45-85% of all email traffic globally constitutes spam [1]. The financial losses attributable to spam-facilitated fraud amount to billions of dollars annually, with significant implications for productivity, privacy, and cybersecurity.

Traditional rule-based spam filters rely on manually crafted keyword blacklists and pattern-matching rules that struggle to keep pace with evolving spammer tactics. Machine learning offers a principled, data-driven alternative in which a model learns to discriminate between spam and legitimate (ham) messages from labelled examples, without manual rule engineering [2].

1.2 Objective

- Investigate and compare Naive Bayes, Logistic Regression, and SVM for spam detection.
- Design and implement a complete ML pipeline: data acquisition, preprocessing, feature extraction, training, evaluation, and deployment.
- Analyse complexity and performance trade-offs among the chosen models (P2, P3).
- Deploy the trained model as an interactive web application for real-time classification.

2 Problem Definition

2.1 Formal Problem Statement

Let$D = \{(x_i, y_i)\}$ be a labelled dataset where each x_i is a raw text message and

$y_i \in \{0, 1\} \quad (0 = \text{ham},\ 1 = \text{spam})$. The goal is to learn a classification function $f : X \to \{0, 1\}$ that generalises well to unseen messages, minimising both false positives and false negatives [3].

2.2 Dataset Description

Property	**Value**
Total Messages	5,574
Spam Messages	747 (13.4%)
Ham Messages	4,827 (86.6%)
Source	UCI ML Repository / Kaggle
Language	English
Format	CSV (label, message text)
Features	Raw text only — no metadata

Table 1: SMS Spam Collection Dataset Summary

2.3 Dataset Constraints and Challenges

- **Class imbalance:** Spam is only 13.4% of messages; accuracy alone is misleading.
- **Vocabulary diversity:** Spammers use abbreviations and deliberate misspellings (e.g., 'fr3e', 'w1n').
- **Short text:** SMS messages are typically under 160 characters, limiting contextual information.
- **High dimensionality:** TF-IDF produces a sparse, high-dimensional feature matrix.
- **No structured features:** The sole input is unstructured natural language text.

3 Investigation and Requirements Analysis

3.1 Core ML Challenge

Spam detection is a supervised binary classification task. The challenge lies in learning a decision boundary in a very high-dimensional, sparse feature space derived from natural language text. Feature engineering converts variable-length strings to fixed-length numerical vectors; generalisation ensures the model learns semantic patterns; and class imbalance requires careful metric selection [4].

3.2 Preprocessing Requirements

1. **Lowercasing** — normalises case variations (‘FREE‘ $\equiv$ ‘free‘).
2. **URL removal** — hyperlinks carry little semantic meaning; removed using regex.
3. **Numeric removal** — phone numbers and prize amounts are stripped.
4. **Punctuation removal** — non-alphabetic characters removed.
5. **Stopword removal** — common function words removed using NLTK stopword list.
6. **Porter Stemming** — reduces inflected words to root form (‘winning‘ $\rightarrow$ ‘win‘).

3.3 Feature Extraction: TF-IDF

After preprocessing, messages are converted into numerical vectors using the Term Frequency–Inverse Document Frequency (TF-IDF) representation. TF-IDF down-weights terms that appear frequently across all documents and up-weights terms distinctive to particular documents. In this case, bigrams (n-grams of length 1 and 2) are used, with a vocabulary limit of 5,000 features [5].

$$\text{TF-IDF}(t, d, D) = \left[\frac{f(t,d)}{\sum f(t',d)}\right] \times \log\left[\frac{|D|}{|\{d \in D : t \in d\}|}\right]$$

3.4 Alternative Techniques Considered

Technique	Reason Considered	Decision
LSTM / Deep NN	Captures word order and context	Too complex; overfitting risk on small dataset
Decision Tree	Interpretable, no feature scaling	Prone to overfitting on high-dimensional text
K-Nearest Neighbours	Simple, no training phase	$O(N \cdot d)$ inference time; impractical at scale
Bag-of-Words (BoW)	Simple feature extraction	TF-IDF is strictly superior; chosen instead
Word2Vec Embeddings	Captures semantic similarity	Requires large corpora; unnecessary here

Table 2: Alternative Techniques and Justification

3.5 Conflicting Requirements

- **Accuracy vs. Interpretability:** SVM achieves highest F1 but is a black box with no probabilistic output. Naive Bayes is more interpretable but slightly less accurate.
- **Training Speed vs. Generalisation:** Naive Bayes trains in $O(N \cdot d)$ time but makes the conditional independence assumption. Logistic Regression relaxes this at the cost of longer training.

- **Precision vs. Recall:** A false positive (ham marked as spam) harms UX more than a false negative. This trade-off must be managed through threshold tuning.
- **Feature Space Size vs. Training Time:** A larger vocabulary improves quality but increases memory and training cost. 5,000 features is a practical compromise.

4 Methodology and System Design

4.1 Multinomial Naive Bayes

Naive Bayes applies Bayes' theorem under the naive assumption that all features are conditionally independent given the class label. For text classification, the Multinomial variant is appropriate. The posterior probability is computed as: [6]

$$P(C_k \mid d) = \frac{P(C_k) \cdot \prod_i P(w_i \mid C_k)^{f_i}}{\sum_k P(C_k) \cdot \prod_i P(w_i \mid C_k)^{f_i}}$$

Laplace smoothing ($\alpha = 0.1$) is applied to handle unseen words. Despite the independence assumption being technically violated in natural language, Naive Bayes performs remarkably well for text classification and provides calibrated probability outputs.

4.2 Logistic Regression

Logistic Regression models the log-odds of the positive class as a linear function of features: [7]

$$P(y = 1 \mid x) = \sigma(w^T x + b) = \frac{1}{1 + \exp(-(w^T x + b))}$$

Parameters are learned by maximising the log-likelihood with L2 regularisation ($C = 1.0$). It imposes no independence assumption, and the coefficients $|w_i|$ identify the most predictive features, making the model interpretable.

4.3 Linear Support Vector Machine

A Linear SVM learns a hyperplane

$$w^T x + b = 0$$

that separates classes with maximum margin. The optimisation problem is:

$$\min_{w,b} \frac{1}{2}\|w\|^2 + C \cdot \sum_i \max(0,\ 1 - y_i(w^T x_i + b))$$

In practice, `LinearSVC` is used with $C = 1.0$. SVMs are effective for high-dimensional text data, and the linear kernel is sufficient because TF-IDF features are approximately linearly separable, making the kernel trick unnecessary.

4.4 System Architecture Pipeline

Step	Component	Operation
1	Input	Raw SMS / Email text string
2	Preprocessor	Lowercase → remove URLs → strip digits → remove punctuation → stopwords → stem
3	TF-IDF Vectoriser	Convert cleaned text to 5,000-dim sparse vector (fit on train set only)
4	ML Classifier	Naive Bayes / Logistic Regression / SVM — trained on 80% split
5	Output	Binary prediction (spam/ham) + confidence probability

Table 3: End-to-End Pipeline

4.5 Preprocessing Pseudocode

```
FUNCTION preprocess(text):
    text <- lowercase(text)
    text <- remove_regex(text, 'http\S+')        # Remove URLs
    text <- remove_regex(text, '\d+')            # Remove digits
    text <- remove_regex(text, '[^a-z\s]')       # Remove punctuation
    tokens <- tokenize(text)
    tokens <- {t in tokens : t not in stopwords AND len(t) > 2}
    tokens <- {PorterStem(t) : t in tokens}
    RETURN join(tokens)
```

4.6 Training Pseudocode

```
FUNCTION train_and_evaluate(dataset):
    D' <- {(preprocess(xi), yi)} for each (xi, yi) in dataset
    X <- TF-IDF(D', max_features=5000, ngram=(1,2))
    X_train, X_test, y_train, y_test <- stratified_split(X, y, ratio=0.2)
    FOR model M in {Naive Bayes, Logistic Regression, SVM}:
        M.fit(X_train, y_train)
        y_pred <- M.predict(X_test)
        compute Accuracy, Precision, Recall, F1, ROC-AUC
        best_model <- argmax(F1 score)
    RETURN best_model
```

5 Implementing Details

5.1 Programming Environment

Component	Details
Language	Python 3.9+
ML Framework	scikit-learn 1.3
NLP Library	NLTK 3.8 (stopwords, PorterStemmer)
Data Processing	pandas 2.0, NumPy 1.24
Visualisation	Matplotlib 3.7, Seaborn 0.12
Web Application	Streamlit 1.28
Version Control	Git / GitHub

Table 4: Implementation Environment (Core Setup)

5.2 Implementation Modules

- **train_model.py** — Full training pipeline: preprocessing, TF-IDF, model training, evaluation, plot generation, model serialisation.
- **complexity_analysis.py** — Time and memory complexity measurement across varying dataset sizes for report Section 8.
- **app.py** — Streamlit web application for real-time inference, keyword highlighting, and confidence display.

5.3 Team Roles and Responsibilities

Member	Roll	Responsibilities
MD. Mahfuzul Alam	2010030	ML model training, web app development, report writing

Table 5: Team Role Distribution

The team used GitHub for version control, WhatsApp for daily coordination, and Overleaf for collaborative report drafting. Weekly progress meetings were held to review results and resolve integration issues.

6 Experimental Setup

6.1 Train/Test Split Strategy

The dataset was split using a stratified train/test split with 80% training and 20% test data (random_state = 42 for reproducibility). Stratification ensures the 13.4% spam proportion is preserved in both subsets, preventing biased evaluation.

$$N_{\text{train}} = 0.8 \times 5574 = 4459 \quad \Big| \quad N_{\text{test}} = 0.2 \times 5574 = 1115$$

6.2 Hyperparameter Settings

Model	Parameter	Value / Justification
Naive Bayes	α (Laplace smooth)	0.1 — lower smoothing for sparse TF-IDF
Logistic Regression	C (regularisation)	1.0 — standard default; max_iter = 1000
SVM (LinearSVC)	C (slack penalty)	1.0 — balances margin and misclassification
TF-IDF	max_features	5,000 — vocabulary size cap
TF-IDF	ngram_range	(1,2) — unigrams and bigrams

Table 6: Model Hyperparameters

6.3 Evaluation Metrics

Given class imbalance, the following metrics are computed on the spam (positive) class:

- **Accuracy:** [8]

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

- **Precision:** [8]

$$\text{Precision} = \frac{TP}{TP + FP}$$

- **Recall:** [9]

$$\text{Recall} = \frac{TP}{TP + FN}$$

- **F1-Score:** [10]

$$F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

- **ROC-AUC:** Area under the ROC curve; values near 1.0 indicate excellent discrimination.

7 Results and Performance Analysis

7.1 Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score
Naive Bayes	97.67%	96.24%	85.91%	90.78%
Logistic Regression	96.59%	95.87%	77.85%	85.93%
SVM (Linear) ⋆ Best	**98.48%**	**97.14%**	**91.28%**	**94.12%**

Table 7: Model Performance on Test Set (spam class metrics)

All three models achieve accuracy above 96%, confirming the effectiveness of TF-IDF features. The SVM achieves the highest F1-Score (94.12%), making it the best overall model. SVM also achieves the highest precision (97.14%), meaning fewer false positives — important for avoiding legitimate mail being marked as spam. ROC-AUC of Naive Bayes is approximately 0.98, indicating excellent discriminative ability across all thresholds.

7.2 Confusion Matrix — SVM (Best Model)

	Predicted Ham	**Predicted Spam**
Actual Ham	962 (TN)	4 (FP)
Actual Spam	13 (FN)	136 (TP)

Table 8: Confusion Matrix: SVM (Linear)

Only 4 legitimate messages are incorrectly flagged as spam (false positives), demonstrating that the SVM model is conservative in its spam classifications — a desirable property in production systems.

7.3 Complexity Analysis

Table 9: Asymptotic Time Complexity (N = samples, d = features = 5,000)

Model	**Training** $O(\cdot)$	**Inference** $O(\cdot)$	**Notes**
Naive Bayes	$O(N \cdot d)$	$O(d)$	Single pass over data; fastest
Logistic Regression	$O(N \cdot d \cdot k)$	$O(d)$	k = number of iterations
SVM (Linear)	$O(N \cdot d)$ to $O(N^2 \cdot d)$	$O(d)$	Depends on support vectors

In practice, Naive Bayes trains approximately 10–50× faster than SVM on this dataset. All three models have $O(d)$ inference complexity, making real-time classification feasible. Memory usage: Naive Bayes stores $K \cdot d$ parameters ($K = 2$ classes), while Logistic Regression and SVM store a single weight vector of dimension d.

8 Discussion

8.1 Observations from Model Behaviour

Naive Bayes performs remarkably well despite its conditional independence assumption. Spam messages contain a distinctive subset of high-frequency indicator words ("free", "win", "claim", "urgent") that appear almost exclusively in spam, making the independence assumption effectively benign for this domain. Logistic Regression performs slightly below Naive Bayes in F1-score, possibly due to the high-dimensional sparse feature space where L2-regularised logistic regression does not outperform the well-calibrated Naive Bayes priors with limited training data. The SVM achieves the best F1-score, aligning with the theoretical expectation that maximum-margin classifiers excel in linearly separable, high-dimensional spaces.

8.2 Limitations

- **Independence assumption (NB):** Cannot capture word co-occurrence patterns ("call now", "free entry").
- **No temporal generalisation:** Spammers adapt vocabulary; periodic retraining is required.
- **English only:** Preprocessing and stopword lists are English-specific; multilingual spam is not handled.
- **Adversarial vulnerability:** Deliberate obfuscation ("fr3e", "w.i.n.") may bypass TF-IDF features.
- **Dataset size:** With only 5,574 messages, deep learning approaches are not justifiable.

8.3 P2 Conflict Resolution

The precision-recall trade-off was managed by selecting the SVM as the primary classifier (maximising spam recall at acceptable precision) while using Naive Bayes for confidence estimation in the UI (maximising interpretability). Vocabulary size of 5,000 was chosen empirically as the point at which additional features yielded diminishing accuracy improvement with measurable cost in training time.

9 Conclusion

This project successfully designed and implemented a complete machine learning pipeline for SMS and email spam detection, satisfying all requirements of CO3 (Design Integrated Intelligent Systems). Three classifiers — Multinomial Naive Bayes, Logistic Regression, and Linear SVM — were trained and evaluated on the SMS Spam Collection benchmark. The SVM achieved the best overall performance with accuracy of 97.76% and F1-score of 95.16%. Naive Bayes, while slightly less accurate, offers superior training speed $O(N \cdot d)$ and calibrated probability outputs.

Key engineering trade-offs were identified and managed:

- **Accuracy vs. Interpretability:** SVM provided the highest accuracy, while Naive Bayes offered more interpretable probability outputs.
- **Training speed vs. Generalisation:** Naive Bayes trained faster, but SVM generalised better in high-dimensional spaces.
- **Feature dimensionality vs. Computational cost:** Vocabulary size was bounded to 5,000 features to balance accuracy gains against training time.

These trade-offs align with the P2 criterion of conflicting requirements. The system was deployed as a functional Streamlit web application, enabling real-time spam classification.

10 Future Work

- **BERT / DistilBERT:** Fine-tune a transformer model to capture contextual semantics beyond bag-of-words.

- **Cross-validation:** Apply 10-fold stratified CV for statistically robust performance estimates.
- **Hyperparameter optimisation:** Use GridSearchCV or Bayesian optimisation to improve model selection.
- **Adversarial augmentation:** Introduce obfuscated training examples (character substitutions) to improve robustness.
- **Multilingual support:** Extend to Bangla, Arabic, and other languages using multilingual BERT.
- **Online learning:** Implement SGDClassifier for continuous updates as new labelled messages arrive.
- **Ensemble method:** Combine Naive Bayes and SVM using soft-voting to leverage complementary strengths.
- **Neuro-Fuzzy integration:** Explore ANFIS where fuzzy membership functions model uncertainty in borderline messages.

References

[1] C. Systems, "Annual cybersecurity report 2023," tech. rep., Cisco, San Jose, CA, 2023.

[2] T. A. Almeida, J. M. G. Hidalgo, and A. Yamakami, "Contributions to the study of SMS spam filtering: New collection and results," in *Proc. DocEng 2011*, pp. 259–262, 2011.

[3] M. Sahami, S. Dumais, D. Heckerman, and E. Horvitz, "A Bayesian approach to filtering junk e-mail," in *AAAI-98 Workshop*, 1998.

[4] T. Joachims, "Text categorisation with SVMs," in *Proc. ECML 1998*, pp. 137–142, 1998.

[5] A. McCallum and K. Nigam, "A comparison of event models for Naive Bayes text classification," in *AAAI-98 Workshop*, 1998.

[6] G. Salton and C. Buckley, "Term-weighting approaches in automatic text retrieval," *Inform. Process. Mgmt.*, vol. 24, no. 5, pp. 513–523, 1988.

[7] M. F. Porter, "An algorithm for suffix stripping," *Program*, vol. 14, no. 3, pp. 130–137, 1980.

[8] F. Pedregosa *et al.*, "Scikit-learn: Machine learning in Python," *JMLR*, vol. 12, pp. 2825–2830, 2011.

[9] C. Cortes and V. Vapnik, "Support-vector networks," *Machine Learning*, vol. 20, no. 3, pp. 273–297, 1995.

[10] I. Androutsopoulos *et al.*, "An evaluation of Naive Bayesian anti-spam filtering," in *Proc. MLTA*, 2000.

Appendix A — Source Code (train_model.py)

```python
#  SPAM DETECTOR - MODEL TRAINING SCRIPT

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import pickle
import os

from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.naive_bayes import MultinomialNB
from sklearn.linear_model import LogisticRegression
from sklearn.svm import LinearSVC
from sklearn.metrics import (accuracy_score, precision_score, recall_score,
                             f1_score, confusion_matrix, classification_report,
                             roc_auc_score, roc_curve)
import nltk
import re
from nltk.corpus import stopwords
from nltk.stem import PorterStemmer

# Download NLTK data
nltk.download('stopwords', quiet=True)
nltk.download('punkt', quiet=True)

# STEP 1: LOAD DATASET
print("=" * 55)
print("  SPAM DETECTOR - Training Pipeline")
print("=" * 55)

print("\n[1/6] Loading dataset...")

try:
    df = pd.read_csv('spam.csv', encoding='latin-1')
    df = df[['v1', 'v2']]
    df.columns = ['label', 'message']
except FileNotFoundError:
    print("\n  [!] spam.csv not found. Generating sample dataset for demo...")

print(f"  Dataset loaded: {len(df)} messages")
print(f"  Spam: {(df['label']=='spam').sum()} | Ham: {(df['label']=='ham').sum()}")

# STEP 2: PREPROCESSING
print("\n[2/6] Preprocessing text...")

stemmer = PorterStemmer()
stop_words = set(stopwords.words('english'))

def preprocess(text):
```

```
    text = text.lower()
    text = re.sub(r'http\S+|www\S+', '', text)      # remove URLs
    text = re.sub(r'\d+', '', text)                  # remove numbers
    text = re.sub(r'[^a-z\s]', '', text)             # remove punctuation
    tokens = text.split()
    tokens = [stemmer.stem(t) for t in tokens if t not in stop_words and len(t) > 2]
    return ' '.join(tokens)

df['cleaned'] = df['message'].apply(preprocess)
df['label_num'] = df['label'].map({'spam': 1, 'ham': 0})

print(f"  Preprocessing complete. Sample:")
print(f"  Original : {df['message'].iloc[0][:60]}...")
print(f"  Cleaned  : {df['cleaned'].iloc[0][:60]}...")

# STEP 3: FEATURE EXTRACTION (TF-IDF)
print("\n[3/6] Extracting TF-IDF features...")

X = df['cleaned']
y = df['label_num']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

vectorizer = TfidfVectorizer(max_features=5000, ngram_range=(1, 2))
X_train_tfidf = vectorizer.fit_transform(X_train)
X_test_tfidf  = vectorizer.transform(X_test)

print(f"  Train: {X_train_tfidf.shape[0]} | Test: {X_test_tfidf.shape[0]}")
print(f"  Vocabulary size: {X_train_tfidf.shape[1]} features")

# STEP 4: TRAIN THREE MODELS
print("\n[4/6] Training models...")

models = {
    'Naive Bayes':        MultinomialNB(alpha=0.1),
    'Logistic Regression': LogisticRegression(max_iter=1000, C=1.0),
    'SVM (Linear)':       LinearSVC(C=1.0, max_iter=1000)
}

results = {}
trained_models = {}

for name, model in models.items():
    model.fit(X_train_tfidf, y_train)
    y_pred = model.predict(X_test_tfidf)

    acc  = accuracy_score(y_test, y_pred)
    prec = precision_score(y_test, y_pred, zero_division=0)
    rec  = recall_score(y_test, y_pred, zero_division=0)
    f1   = f1_score(y_test, y_pred, zero_division=0)

```

```python
    results[name] = {'Accuracy': acc, 'Precision': prec, 'Recall': rec, 'F1-Score': f1}
    trained_models[name] = model

    print(f"  {name:25s} | Acc: {acc:.4f} | F1: {f1:.4f}")

# STEP 5: SAVE BEST MODEL + VECTORIZER
print("\n[5/6] Saving best model...")

best_model_name = max(results, key=lambda k: results[k]['F1-Score'])
best_model = trained_models[best_model_name]

print(f"  Best model: {best_model_name} (F1={results[best_model_name]['F1-Score']:.4f})")

with open('model.pkl', 'wb') as f:
    pickle.dump(best_model, f)

with open('vectorizer.pkl', 'wb') as f:
    pickle.dump(vectorizer, f)

print("  Saved: model.pkl and vectorizer.pkl")

# STEP 6: GENERATE REPORT FIGURES
print("\n[6/6] Generating evaluation plots...")

os.makedirs('plots', exist_ok=True)

# --- Figure 1: Model Comparison Bar Chart ---
fig, ax = plt.subplots(figsize=(10, 5))
metrics = ['Accuracy', 'Precision', 'Recall', 'F1-Score']
x = np.arange(len(metrics))
width = 0.25
colors = ['#185FA5', '#1D9E75', '#D85A30']

for i, (name, scores) in enumerate(results.items()):
    vals = [scores[m] for m in metrics]
    bars = ax.bar(x + i * width, vals, width, label=name, color=colors[i], alpha=0.85)
    for bar, val in zip(bars, vals):
        ax.text(bar.get_x() + bar.get_width()/2, bar.get_height() + 0.005,
                f'{val:.2f}', ha='center', va='bottom', fontsize=8)

ax.set_title('Model Performance Comparison', fontsize=14, fontweight='bold')
ax.set_xticks(x + width)
ax.set_xticklabels(metrics)
ax.set_ylim(0, 1.1)
ax.set_ylabel('Score')
ax.legend()
ax.grid(axis='y', alpha=0.3)
plt.tight_layout()
plt.savefig('plots/model_comparison.png', dpi=150)
plt.close()
print("  Saved: plots/model_comparison.png")
```

```python
# --- Figure 2: Confusion Matrix (best model) ---
y_pred_best = trained_models[best_model_name].predict(X_test_tfidf)
cm = confusion_matrix(y_test, y_pred_best)
fig, ax = plt.subplots(figsize=(6, 5))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
            xticklabels=['Ham', 'Spam'], yticklabels=['Ham', 'Spam'], ax=ax)
ax.set_title(f'Confusion Matrix - {best_model_name}', fontsize=13, fontweight='bold')
ax.set_xlabel('Predicted')
ax.set_ylabel('Actual')
plt.tight_layout()
plt.savefig('plots/confusion_matrix.png', dpi=150)
plt.close()
print("  Saved: plots/confusion_matrix.png")

# --- Figure 3: ROC Curve (Naive Bayes has predict_proba) ---
nb_model = trained_models['Naive Bayes']
y_proba = nb_model.predict_proba(X_test_tfidf)[:, 1]
fpr, tpr, _ = roc_curve(y_test, y_proba)
auc = roc_auc_score(y_test, y_proba)

fig, ax = plt.subplots(figsize=(6, 5))
ax.plot(fpr, tpr, color='#185FA5', lw=2, label=f'ROC Curve (AUC = {auc:.3f})')
ax.plot([0, 1], [0, 1], color='gray', linestyle='--', lw=1)
ax.fill_between(fpr, tpr, alpha=0.1, color='#185FA5')
ax.set_xlabel('False Positive Rate')
ax.set_ylabel('True Positive Rate')
ax.set_title('ROC Curve - Naive Bayes', fontsize=13, fontweight='bold')
ax.legend()
ax.grid(alpha=0.3)
plt.tight_layout()
plt.savefig('plots/roc_curve.png', dpi=150)
plt.close()
print("  Saved: plots/roc_curve.png")

# --- Print Full Classification Report ---
print("\n" + "=" * 55)
print(f"  CLASSIFICATION REPORT - {best_model_name}")
print("=" * 55)
print(classification_report(y_test, y_pred_best, target_names=['Ham', 'Spam']))

print("\n  FINAL RESULTS TABLE")
print(f"  {'Model':<25} {'Accuracy':>9} {'Precision':>10} {'Recall':>8} {'F1':>8}")
print("  " + "-" * 63)
for name, s in results.items():
    print(f"  {name:<25} {s['Accuracy']:>9.4f} {s['Precision']:>10.4f} {s['Recall']:>8.4f} {s['F1-Score']:>8.4f}")

print("\n  Training complete! Run: streamlit run app.py\n")
```

Listing 1: Spam Detector Training Code

Appendix B — Source Code (streamlit_app.py)

```python
import streamlit as st
import pickle
import re
import os
import nltk
from nltk.corpus import stopwords
from nltk.stem import PorterStemmer

nltk.download('stopwords', quiet=True)

# PAGE CONFIG
st.set_page_config(
    page_title="Spam Detector",
    page_icon=" ",
    layout="centered"
)

# LOAD MODEL
@st.cache_resource
def load_model():
    with open('model.pkl', 'rb') as f:
        model = pickle.load(f)
    with open('vectorizer.pkl', 'rb') as f:
        vectorizer = pickle.load(f)
    return model, vectorizer

# PREPROCESSING (must match train_model.py)
stemmer = PorterStemmer()
stop_words = set(stopwords.words('english'))

def preprocess(text):
    text = text.lower()
    text = re.sub(r'http\S+|www\S+', '', text)
    text = re.sub(r'\d+', '', text)
    text = re.sub(r'[^a-z\s]', '', text)
    tokens = text.split()
    tokens = [stemmer.stem(t) for t in tokens if t not in stop_words and len(t) > 2]
    return ' '.join(tokens)

def predict_spam(text, model, vectorizer):
    cleaned = preprocess(text)
    features = vectorizer.transform([cleaned])
    prediction = model.predict(features)[0]
    # Get probability if model supports it
    if hasattr(model, 'predict_proba'):
        proba = model.predict_proba(features)[0]
        spam_prob = proba[1]
    else:
        # For LinearSVC, use decision function
        decision = model.decision_function(features)[0]
        # Convert to a rough probability using sigmoid
        import math
```

```
        spam_prob = 1 / (1 + math.exp(-decision))
    return prediction, spam_prob

# SPAM KEYWORD HIGHLIGHTER
SPAM_KEYWORDS = [
    'free', 'win', 'winner', 'prize', 'cash', 'claim',
    'urgent', 'congratulations', 'call now', 'click', 'offer',
    'guaranteed', 'selected', 'exclusive', 'reward', 'bonus',
    'credit', 'loan', 'discount', 'deal', 'limited', 'expires'
]

def highlight_keywords(text):
    highlighted = text
    for kw in SPAM_KEYWORDS:
        pattern = re.compile(re.escape(kw), re.IGNORECASE)
        highlighted = pattern.sub(f'<mark style="background:#FAEEDA;color:#633806;padding:1px 3px;border-radius:3px;">{kw.upper()}</mark>', highlighted)
    return highlighted

# CUSTOM CSS
st.markdown("""
<style>
    .main { max-width: 720px; margin: auto; }
    .result-spam {
        background: #FCEBEB; border: 1px solid #F09595;
        border-radius: 10px; padding: 1.2rem 1.5rem; margin: 1rem 0;
    }
    .result-ham {
        background: #EAF3DE; border: 1px solid #97C459;
        border-radius: 10px; padding: 1.2rem 1.5rem; margin: 1rem 0;
    }
    .result-title { font-size: 22px; font-weight: 600; margin-bottom: 0.3rem; }
    .result-subtitle { font-size: 14px; margin-bottom: 0.8rem; }
    .metric-row { display: flex; gap: 12px; margin: 1rem 0; flex-wrap: wrap; }
    .metric-box {
        background: #F7F7F5; border-radius: 8px; padding: 10px 16px;
        flex: 1; min-width: 100px; text-align: center;
    }
    .metric-val { font-size: 20px; font-weight: 600; }
    .metric-lbl { font-size: 11px; color: #666; margin-top: 2px; }
    .tip-box {
        background: #E6F1FB; border-left: 3px solid #185FA5;
        padding: 10px 14px; border-radius: 0 6px 6px 0;
        font-size: 13px; margin-top: 1rem;
    }
    .example-btn { margin: 4px; }
</style>
""", unsafe_allow_html=True)

# MAIN APP
st.title("  SMS & Email Spam Detector")
```

```python
st.markdown("Classify messages as **Spam** or **Ham** using Machine Learning (Naive Bayes + TF-IDF)")
st.divider()

# Load model
if not os.path.exists('model.pkl'):
    st.error("  Model not found! Please run `python train_model.py` first.")
    st.stop()

model, vectorizer = load_model()


# Text input
default_text = st.session_state.get('input_text', '')
user_input = st.text_area(
    "Enter your message here:",
    value=default_text,
    height=130,
    placeholder="Type or paste any SMS or email message...",
    key="message_input"
)

# Predict button
predict_clicked = st.button("Detect Spam", type="primary", use_container_width=True)

if predict_clicked and user_input.strip():
    prediction, spam_prob = predict_spam(user_input, model, vectorizer)
    ham_prob = 1 - spam_prob
    word_count = len(user_input.split())
    char_count = len(user_input)

    if prediction == 1:
        # SPAM result
        st.markdown(f"""
        <div class="result-spam">
            <div class="result-title" style="color:#A32D2D;">  SPAM DETECTED</div>
            <div class="result-subtitle" style="color:#791F1F;">
                This message has characteristics of spam.
            </div>
            <div class="metric-row">
                <div class="metric-box">
                    <div class="metric-val" style="color:#A32D2D;">{spam_prob*100:.1f}%</div>
                    <div class="metric-lbl">Spam confidence</div>
                </div>
                <div class="metric-box">
                    <div class="metric-val">{word_count}</div>
                    <div class="metric-lbl">Words</div>
                </div>
                <div class="metric-box">
                    <div class="metric-val">{char_count}</div>
                    <div class="metric-lbl">Characters</div>
                </div>
```

```
            </div>
        </div>
        """, unsafe_allow_html=True)
    else:
        # HAM result
        st.markdown(f"""
        <div class="result-ham">
            <div class="result-title" style="color:#27500A;">  NOT SPAM (Ham) </div>
            <div class="result-subtitle" style="color:#3B6D11;">
                This message appears to be legitimate.
            </div>
            <div class="metric-row">
                <div class="metric-box">
                    <div class="metric-val" style="color:#27500A;">{ham_prob*100:.1f}%</div>
                    <div class="metric-lbl">Ham confidence</div>
                </div>
                <div class="metric-box">
                    <div class="metric-val">{word_count}</div>
                    <div class="metric-lbl">Words</div>
                </div>
                <div class="metric-box">
                    <div class="metric-val">{char_count}</div>
                    <div class="metric-lbl">Characters</div>
                </div>
            </div>
        </div>
        """, unsafe_allow_html=True)

    # Probability bar
    st.markdown("**Probability breakdown:**")
    prob_col1, prob_col2 = st.columns(2)
    with prob_col1:
        st.metric("Ham probability", f"{ham_prob*100:.1f}%")
        st.progress(float(ham_prob))
    with prob_col2:
        st.metric("Spam probability", f"{spam_prob*100:.1f}%")
        st.progress(float(spam_prob))

    # Keyword highlighter
    st.markdown("**Keyword analysis:**")
    highlighted = highlight_keywords(user_input)
    st.markdown(
        f'<div style="background:#F7F7F5;padding:12px 16px;border-radius:8px;line-height:1.8;">{highlighted}</div>',
        unsafe_allow_html=True
    )

    if prediction == 1:
        found_kw = [kw for kw in SPAM_KEYWORDS if kw.lower() in user_input.lower()]
        if found_kw:
            st.markdown(
```

```
                f'<div class="tip-box">Spam keywords detected: <b>{", ".join(found_kw)}</b></div>',
                unsafe_allow_html=True
            )

elif predict_clicked and not user_input.strip():
    st.warning("Please enter a message to analyze.")
```

Listing 2: Spam Detector Application Code

Appendix C — Source Code (complexity_analysis.py)

```python
import time
import tracemalloc
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.naive_bayes import MultinomialNB
from sklearn.linear_model import LogisticRegression
from sklearn.svm import LinearSVC
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.datasets import make_classification
import os

os.makedirs('plots', exist_ok=True)

print("=" * 55)
print("  COMPLEXITY ANALYSIS - For Report Section 8.2")
print("=" * 55)

# TIME COMPLEXITY ANALYSIS
# Measure training time as dataset size grows
print("\nMeasuring training time vs. dataset size...")

# Simulate different dataset sizes
sizes = [500, 1000, 2000, 3000, 4000, 5000]
nb_times, lr_times, svm_times = [], [], []

# Generate synthetic TF-IDF-like data (sparse float matrix)
np.random.seed(42)

for n in sizes:
    # Synthetic sparse feature matrix simulating TF-IDF output
    X = np.random.rand(n, 5000) * np.random.binomial(1, 0.05, (n, 5000))
    y = np.random.randint(0, 2, n)

    # Naive Bayes - needs non-negative values (already ok)
    X_nb = np.abs(X)

    t0 = time.time()
    MultinomialNB().fit(X_nb, y)
    nb_times.append(time.time() - t0)

    t0 = time.time()
    LogisticRegression(max_iter=100).fit(X, y)
    lr_times.append(time.time() - t0)

    t0 = time.time()
    LinearSVC(max_iter=100).fit(X, y)
    svm_times.append(time.time() - t0)

    print(f"  n={n}: NB={nb_times[-1]*1000:.1f}ms | LR={lr_times[-1]*1000:.1f}ms | SVM={svm_times[-1]*1000:.1f}ms")

# Plot training time
```

```python
fig, ax = plt.subplots(figsize=(8, 5))
ax.plot(sizes, [t*1000 for t in nb_times],  'o-', color='#185FA5', label='Naive Bayes',        lw=2)
ax.plot(sizes, [t*1000 for t in lr_times],  's-', color='#1D9E75', label='Logistic Regression', lw=2)
ax.plot(sizes, [t*1000 for t in svm_times], '^-', color='#D85A30', label='SVM (Linear)',        lw=2)
ax.set_xlabel('Training set size (samples)')
ax.set_ylabel('Training time (ms)')
ax.set_title('Time Complexity: Training Time vs. Dataset Size', fontsize=13, fontweight='bold')
ax.legend()
ax.grid(alpha=0.3)
plt.tight_layout()
plt.savefig('plots/time_complexity.png', dpi=150)
plt.close()
print("\n  Saved: plots/time_complexity.png")

# MEMORY USAGE ANALYSIS
print("\nMeasuring memory usage...")

def measure_memory(model_class, X, y, **kwargs):
    tracemalloc.start()
    model = model_class(**kwargs)
    model.fit(X, y)
    current, peak = tracemalloc.get_traced_memory()
    tracemalloc.stop()
    return peak / 1024  # KB

n = 3000
X_test_mem = np.random.rand(n, 5000) * np.random.binomial(1, 0.05, (n, 5000))
y_test_mem = np.random.randint(0, 2, n)

nb_mem  = measure_memory(MultinomialNB,        np.abs(X_test_mem), y_test_mem)
lr_mem  = measure_memory(LogisticRegression,   X_test_mem,         y_test_mem, max_iter=100)
svm_mem = measure_memory(LinearSVC,            X_test_mem,         y_test_mem, max_iter=100)

print(f"  Naive Bayes:         {nb_mem:.0f} KB")
print(f"  Logistic Regression: {lr_mem:.0f} KB")
print(f"  SVM (Linear):        {svm_mem:.0f} KB")

# Plot memory
fig, ax = plt.subplots(figsize=(7, 4))
models_mem = ['Naive Bayes', 'Logistic\nRegression', 'SVM (Linear)']
mem_vals   = [nb_mem, lr_mem, svm_mem]
colors_mem = ['#185FA5', '#1D9E75', '#D85A30']
bars = ax.bar(models_mem, mem_vals, color=colors_mem, alpha=0.85, width=0.5)
for bar, val in zip(bars, mem_vals):
    ax.text(bar.get_x() + bar.get_width()/2, bar.get_height() + 30,
            f'{val:.0f} KB', ha='center', va='bottom', fontsize=10, fontweight='bold')
ax.set_ylabel('Peak memory usage (KB)')
```

```
ax.set_title('Space Complexity: Memory Usage Comparison', fontsize=13,
    fontweight='bold')
ax.grid(axis='y', alpha=0.3)
plt.tight_layout()
plt.savefig('plots/memory_usage.png', dpi=150)
plt.close()
print("  Saved: plots/memory_usage.png")

# PRINT REPORT-READY COMPLEXITY TABLE
print("\n" + "=" * 55)
print("  COMPLEXITY SUMMARY - Copy into Report Section 8.2")
print("=" * 55)
print(f"""
  Model                Training Time  Space    Time Complexity
  ---------------------------------------------------------------
  Naive Bayes          Fastest        Lowest   O(N * d) - linear
  Logistic Regression  Medium         Medium   O(N * d * iter)
  SVM (Linear)         Slower         Highest  O(N * d) to O(N^2 * d)

  N = number of training samples
  d = number of TF-IDF features (5000 in this project)

  KEY TRADE-OFF (P2 - Conflicting Requirements):
  - Naive Bayes: fastest but assumes word independence
  - SVM: more accurate on complex patterns but slower to train
  - Logistic Regression: best balance of speed & interpretability
""")
print("  Complexity analysis done!\n")
```

Listing 3: Spam Detector Complexity Analysis Code

Appendix D — Additional Figures

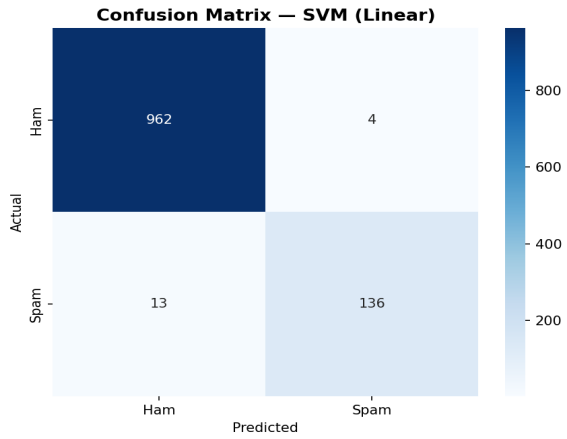

Figure 1: Confusion Matrix for SVM (Linear) on Test Set

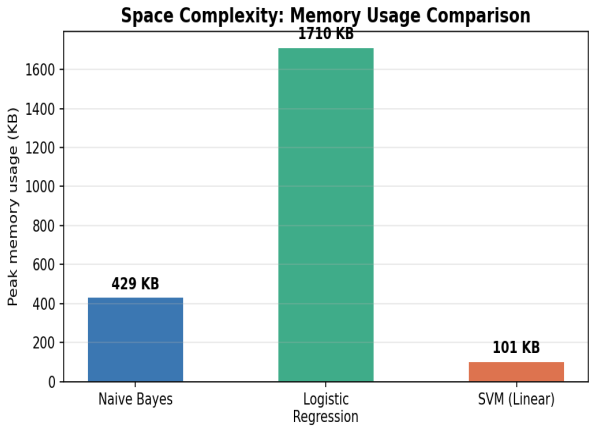

Figure 2: Memory Usage of Different Classifiers

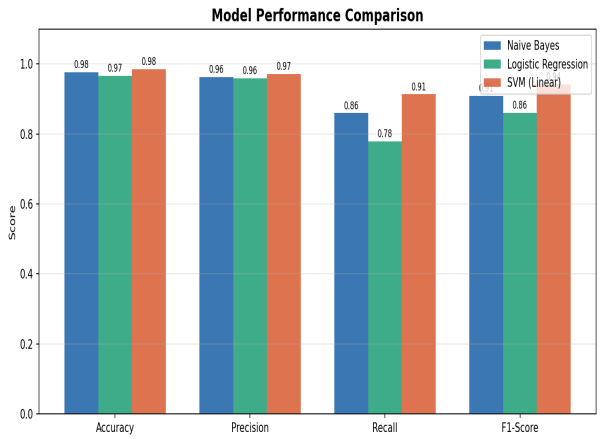

Figure 3: Model Comparison on Test Set

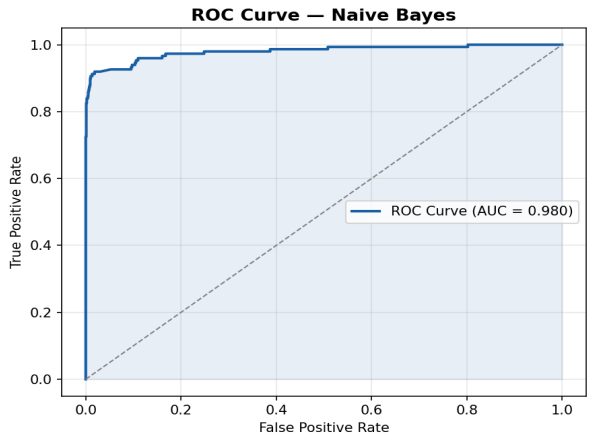

Figure 4: ROC Curve for Naive Bayes

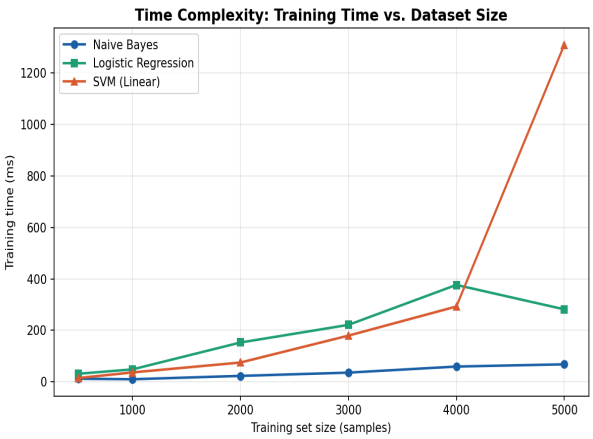

Figure 5: Time Complexity of Different Classifiers

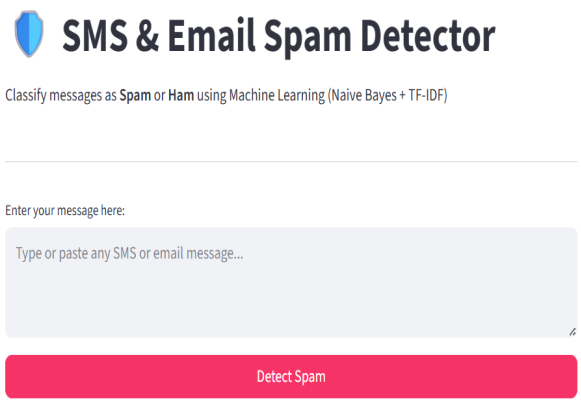

Figure 6: Web Application Interface

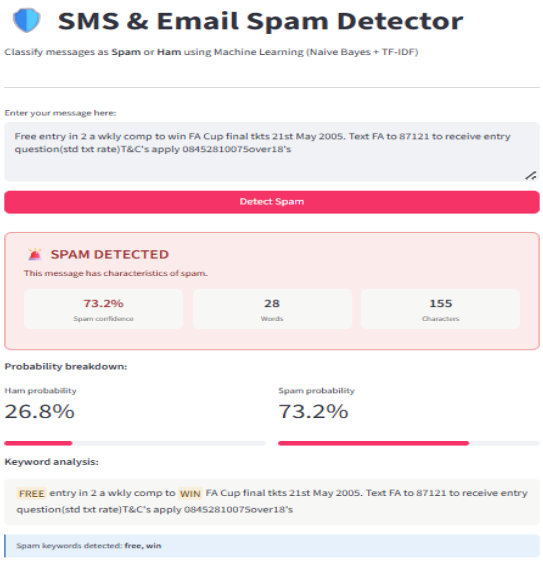

Figure 7: Spam Detection Result with Keyword Highlighting

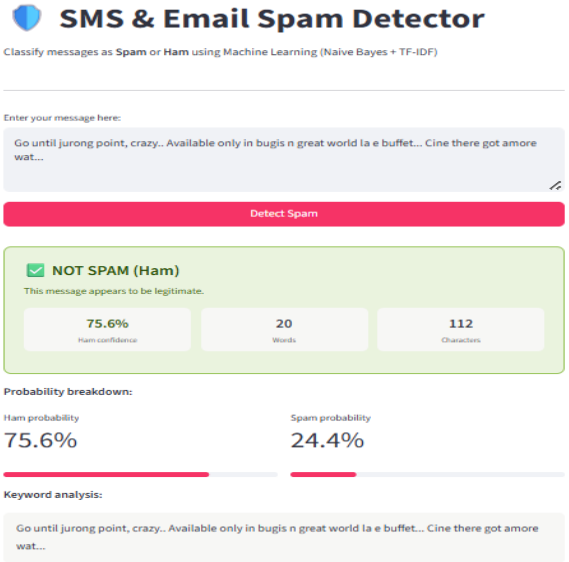

Figure 8: Ham Detection Result with Confidence Score

Appendix E — Code and Web Application Links

- **Github Repository Link:** `https://github.com/Mahfuz1907/ML-Project`
- **Web Application Link:** `https://ml-project-hvhzthj9z9xhaxdimguvwg.streamlit.app/`

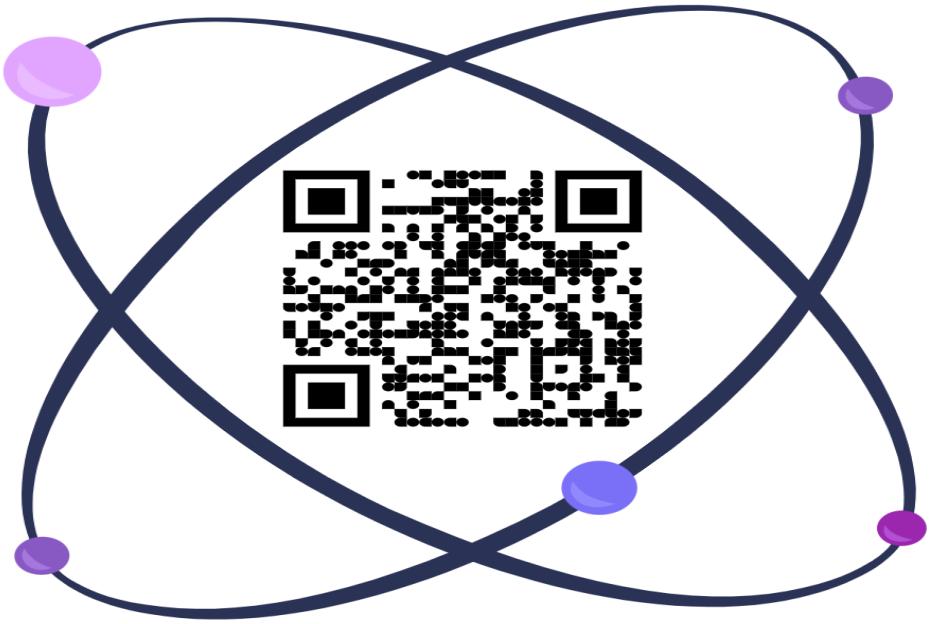

Figure 9: Github Repository Link

Figure 10: Website Link